

Traffic Control System for Metropolitan City

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Labs: Control Traffic System

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Introduction

This project simulates a smart traffic control system for a metropolitan city. It manages road intersections, signal timings, vehicle counts, congestion, emergency vehicles, and accidents.

Roman Urdu: Ye project aik smart traffic control system ko simulate karta hai jo shehar ke intersections, signals, vehicles, traffic jam, emergency gaariyon aur accidents ko manage karta hai.

Objectives

Manage up to 100 intersections using arrays. Update traffic lights dynamically. Detect heavy traffic when vehicles exceed 100. Give green signal to ambulance, police, and fire brigade. Generate daily traffic reports.

Roman Urdu: Is project ka maqsad 100 tak intersections ko arrays mein store karna, signals update karna, traffic jam detect karna, emergency gaariyon ko green signal dena aur daily reports banana hai.

Basic C++ Concepts Used

Variables, constants, cin/cout, arithmetic operators, if - else, switch, loops, functions, arrays, and optional file handling.

Roman Urdu: Is project mein variables, constants, input/output, operators, if - else, switch, loops, functions, arrays aur file handling use ki gayi hai.

Main Menu Options

Add Intersection, Display Intersections, Update Signal Timing, Count Vehicles, Detect Congestion, Emergency Priority, Accident Management, Daily Report, Search, Delete, Save,

Load, Exit.

Roman Urdu: Main menu ke zariye user mukhtalif options select karke system ko control kar sakta hai.

Data Stored for Each Intersection

Intersection Name, Number of Roads, Signal Status, Red Time, Yellow Time, Green Time, Vehicle Count, Congestion Status.

Roman Urdu: Har intersection ka naam, roads ki tadaad, signal status, timings, vehicle count aur congestion status store hota hai.

Important Functions

`addIntersection()`, `displayIntersections()`, `updateSignalTiming()`, `countVehicles()`, `detectCongestion()`, `emergencyPriority()`, `accidentManagement()`, `dailyReport()`.

Roman Urdu: Har kaam ke liye alag function banaya gaya hai taake program ko samajhna aur manage karna asaan ho.

Congestion Detection Logic

If total vehicles are greater than 100, the system displays Heavy Traffic; otherwise Normal Traffic.

Roman Urdu: Agar total vehicles 100 se zyada hon to Heavy Traffic show hota hai, warna Normal Traffic.

Emergency Vehicle Priority

Ambulance, police, and fire brigade receive an immediate green signal.

Roman Urdu: Ambulance, police aur fire brigade ke liye foran green signal activate ho jata hai.

Accident Management

The system records the accident location, blocks the route, and suggests an alternative route.

Roman Urdu: Accident ki surat mein system location save karta hai, route block karta hai aur doosra rasta suggest karta hai.

Daily Report

Displays total vehicles processed, number of congested intersections, emergency events, and accidents.

Roman Urdu: Daily report mein total vehicles, traffic jams, emergencies aur accidents ki details hoti hain.

Advantages

Improves traffic flow, reduces waiting time, prioritizes emergency services, and helps city management.

Roman Urdu: Ye system traffic ko behtar banata hai, waiting time kam karta hai aur emergency services ko priority deta hai.

Conclusion

This beginner - friendly C++ project demonstrates how programming fundamentals can solve real - world traffic problems.

Roman Urdu: Ye beginners ke liye behtareen C++ project hai jo real - life traffic masail ko solve karta hai.

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Complete C++ Program with Line - by - Line Comments

```
#include <iostream>    // Input aur output ke liye library
using namespace std;   // Standard namespace use karne ke liye

int totalVehicles = 0; // Total vehicles ko store karne ke liye global variable

// Vehicles count karne wala function
void countVehicles()
{
    int cars, buses, bikes, trucks; // Mukhtalif vehicle types

    cout << "Enter number of cars: ";
    cin >> cars;

    cout << "Enter number of buses: ";
    cin >> buses;

    cout << "Enter number of bikes: ";
    cin >> bikes;

    cout << "Enter number of trucks: ";
    cin >> trucks;

    totalVehicles = cars + buses + bikes + trucks; // Total calculate karna

    cout << "Total Vehicles = " << totalVehicles << endl;
}

// Traffic congestion detect karne wala function
void detectCongestion()
{
    if (totalVehicles > 100) // Agar vehicles 100 se zyada hon
    {
        cout << "Heavy Traffic Detected!" << endl;
    }
    else
    {
        cout << "Normal Traffic." << endl;
    }
}

// Emergency vehicle ko priority dene wala function
```

```

void emergencyPriority()
{
    int choice;

    cout << "1. Ambulance" << endl;
    cout << "2. Police" << endl;
    cout << "3. Fire Brigade" << endl;
    cout << "Enter emergency vehicle type: ";
    cin >> choice;

    if (choice >= 1 && choice <= 3)
    {
        cout << "Green Signal Activated for Emergency Vehicle!" << endl;
    }
    else
    {
        cout << "Invalid Choice!" << endl;
    }
}

// Main function jahan se program start hota hai
int main()
{
    int option; // Menu choice store karne ke liye

    do
    {
        cout << "\n===== Traffic Control System =====" << endl;
        cout << "1. Count Vehicles" << endl;
        cout << "2. Detect Congestion" << endl;
        cout << "3. Emergency Priority" << endl;
        cout << "4. Exit" << endl;
        cout << "Enter your choice: ";
        cin >> option;

        switch(option)
        {
            case 1:
                countVehicles(); // Vehicles count karna
                break;

            case 2:
                detectCongestion(); // Traffic jam check karna

```

```
        break;

    case 3:
        emergencyPriority(); // Emergency ko priority dena
        break;

    case 4:
        cout << "Program Ended." << endl;
        break;

    default:
        cout << "Invalid Choice!" << endl;
    }

} while(option != 4); // Jab tak user Exit select na kare

return 0; // Program successfully khatam
}
```

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